Celebal Technologies Virtual Internship

Final Project Report

Retail Intelligence using Databricks and PySpark

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B.Tech Information Technology

Academic Session:2026-2027

Internship Duration:

19 May 2026 to 19July 2026

Project Mentor:

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Raj Sir

Internship Overview

During this internship at **Celebal Technologies**, I learned the fundamentals of modern data engineering by completing weekly assignments and finally implementing a complete end-to-end retail analytics pipeline using Databricks and PySpark.

The internship was designed in a progressive manner. Every week's learning became the foundation for the following week's assignment. The concepts gradually moved from Python programming to distributed data processing and finally to building a real-world data engineering project.

The final project brought together all the concepts learned throughout the internship, including ETL pipelines, data cleaning, PySpark transformations, Delta Lake, Medallion Architecture, Star Schema modelling and business KPI generation.

Weekly Learning Journey

Week	
Week 1	Python Programming Fundamentals
Week 2	Data Analysis using Pandas
Week 3	SQL Queries and Database Operations
Week 4	Data Engineering Concepts, ETL Pipelines and Azure Fundamentals
Week 5	Data Cleaning and PySpark
Week 6	Apache Spark Fundamentals
Week 7	Databricks Platform
Week 8	Retail Intelligence Mini Project

Week 1 – Python

During the first week, I revised Python programming concepts such as loops, functions, lists, dictionaries and file handling. These concepts later helped me understand PySpark programming.

Week 2 – Pandas

I learned how to analyse CSV datasets using Pandas.

Some important concepts included:

- Reading CSV files
- Handling missing values
- Filtering records
- Creating new columns
- Performing aggregations
- Exporting cleaned datasets

Week 3 – SQL

During this week I practiced writing SQL queries for analysing structured data.

The topics covered included:

- SELECT
- WHERE
- GROUP BY
- HAVING
- ORDER BY
- Aggregate Functions
- Different Types of Joins

Week 4 – Data Engineering Concepts

This week introduced the overall workflow followed in modern data engineering projects.

Some important concepts learned were:

- ETL Pipeline
- ELT Pipeline
- Data Lake
- Data Warehouse
- Azure Fundamentals
- Batch Processing

Week 5 – PySpark

This week focused on handling large datasets using PySpark.

Topics learned:

- Spark DataFrames
- Transformations
- Actions
- Data Cleaning
- Filtering
- Aggregations
- Column Operations

Week 6 – Apache Spark

This week helped me understand how Spark processes big data efficiently.

Concepts learned:

- Spark Architecture
- Driver and Executors
- Partitions
- Lazy Evaluation
- Distributed Processing

Week 7 – Databricks

I learned how Databricks simplifies data engineering workflows by integrating Apache Spark with cloud storage and collaborative notebooks.

Topics covered:

- Workspace
- Volumes
- Catalog
- Delta Tables
- Notebook Development
- Data Pipeline Execution

Week 8 –Mini Project

The final week involved building a complete Retail Intelligence project by combining all the concepts learned throughout the internship.

```
(venv) himanshu@Himanshus-MacBook-Air-2 ecommerce-analytics-system % python scripts/report_cli.py --report segments

E-COMMERCE ANALYTICS REPORT
Report: SEGMENTS
```

customer_id	customer_name	total_orders	frequency_segment
56	Emily Fischer	9	LOYAL
39	Tony Vazquez	6	LOYAL
92	Lindsey Johnson	6	LOYAL
332	Brandon Shaffer	6	LOYAL
348	Holly Scott	6	LOYAL
391	Deborah Campbell	6	LOYAL
572	Mrs. Melanie Collins	6	LOYAL
588	Danielle Garrett	6	LOYAL
13	James Ferrell	5	LOYAL
42	Sierra Johnson	5	LOYAL
80	Pamela Lopez	5	LOYAL
93	Phillip Underwood	5	LOYAL
122	Mark Watson	5	LOYAL
136	Michelle James	5	LOYAL
152	Donna Sullivan	5	LOYAL
206	Kenneth Montgomery	5	LOYAL
211	Cheryl Moore	5	LOYAL
212	Kathy Levy	5	LOYAL
223	Kaitlyn Charles	5	LOYAL
309	Lisa Cervantes	5	LOYAL

```
(venv) himanshu@Himanshus-MacBook-Air-2 ecommerce-analytics-system % python scripts/report_cli.py --report top_customers

E-COMMERCE ANALYTICS REPORT
Report: TOP_CUSTOMERS
```

customer_id	customer_name	lifetime_value
93	Phillip Underwood	1564726.71
56	Emily Fischer	1307642.38
77	Peter Vaughn Dds	1299837.77
572	Mrs. Melanie Collins	1273559.07
340	Dennis Williams	1219958.24
384	Cynthia Haas	1207642.41
211	Cheryl Moore	1198829.8
117	Amanda Diaz	1186752.38
320	Donna Adams	1161185.95
445	Mariah Robles	1023861.33

```
(venv) himanshu@Himanshus-MacBook-Air-2 ecommerce-analytics-system % python scripts/report_cli.py --report revenue

E-COMMERCE ANALYTICS REPORT
Report: REVENUE
```

category	total_revenue
Books	40727157.81
Home	39255381.55
Clothing	36619976.67
Electronics	34357325.76


```
(venv) himanshu@Himanshus-MacBook-Air-2 ecommerce-analytics-system % python scripts/report_cli.py --report segments

E-COMMERCE ANALYTICS REPORT
Report: SEGMENTS
```

customer_id	customer_name	total_orders	frequency_segment
56	Emily Fischer	9	LOYAL
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206	Kenneth Montgomery	5	LOYAL
211	Cheryl Moore	5	LOYAL
212	Kathy Levy	5	LOYAL
223	Kaitlyn Charles	5	LOYAL
309	Lisa Cervantes	5	LOYAL

Retail Intelligence Project Overview

The objective of this project was to build a complete batch data pipeline capable of processing retail data from multiple sources and generating business-ready analytical tables.

The project uses three datasets:

- Customer Dataset
- Product Dataset
- Sales Dataset

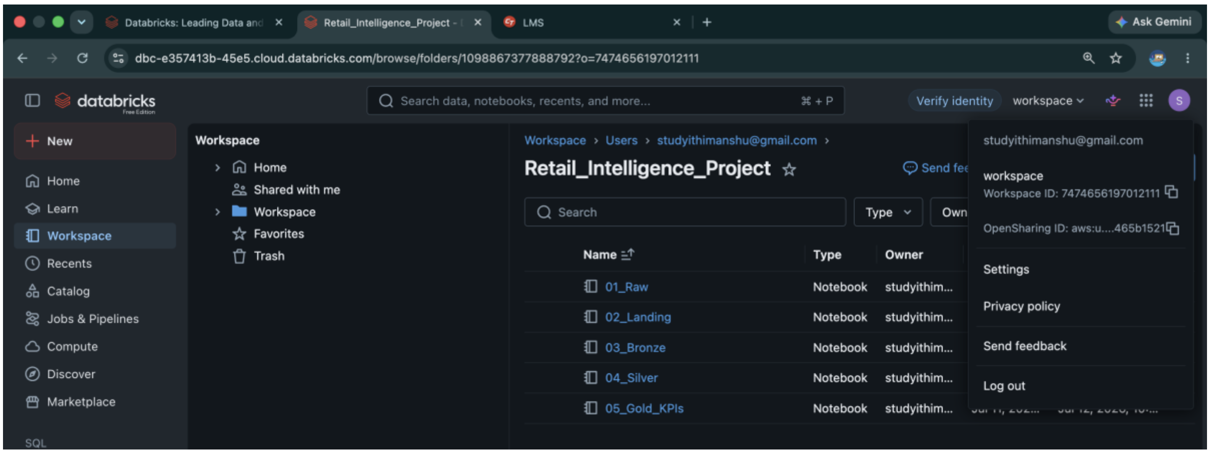
Each dataset contains both:

- Historical Data
- Incremental Data

Historical files represent the existing business records, while incremental files simulate newly arriving business data. This approach helps demonstrate incremental data processing, which is commonly used in real-world data engineering systems.

The project was developed using the Medallion Architecture, where the data moves through multiple layers before becoming ready for business reporting.

Project Folder Structure



(Databricks Workspace Screenshot)

Medallion Architecture

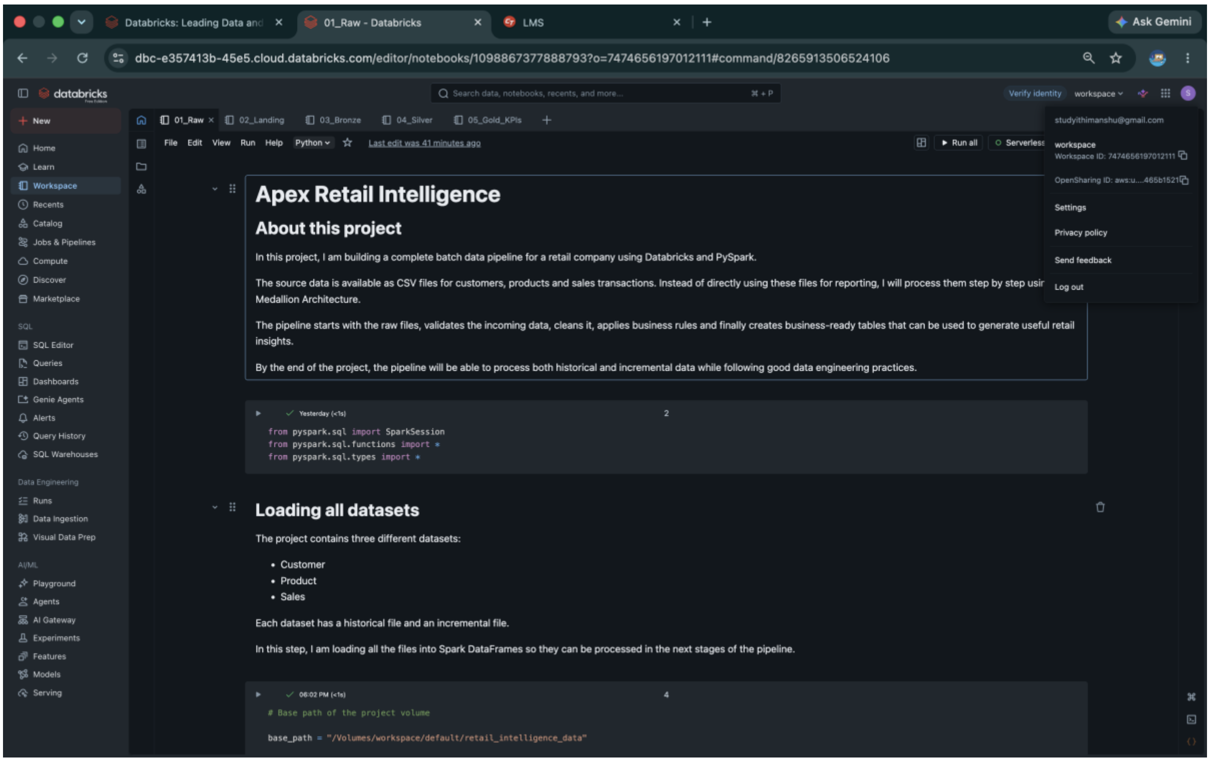
The project follows the Medallion Architecture, which divides data processing into multiple layers.

Raw Layer

The Raw layer stores the original CSV files without making any modifications.

Purpose:

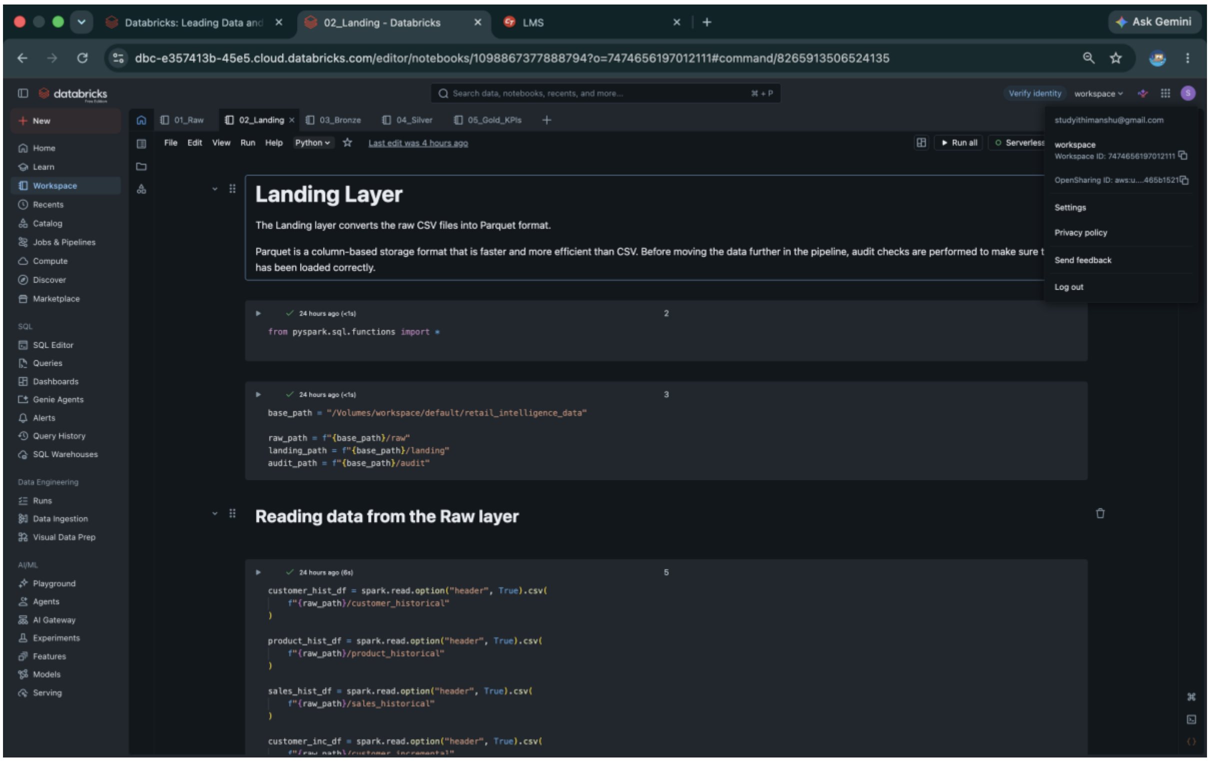
- Preserve original data
- Create backup
- Source for future processing



Landing Layer

The Landing layer converts CSV files into Parquet format.

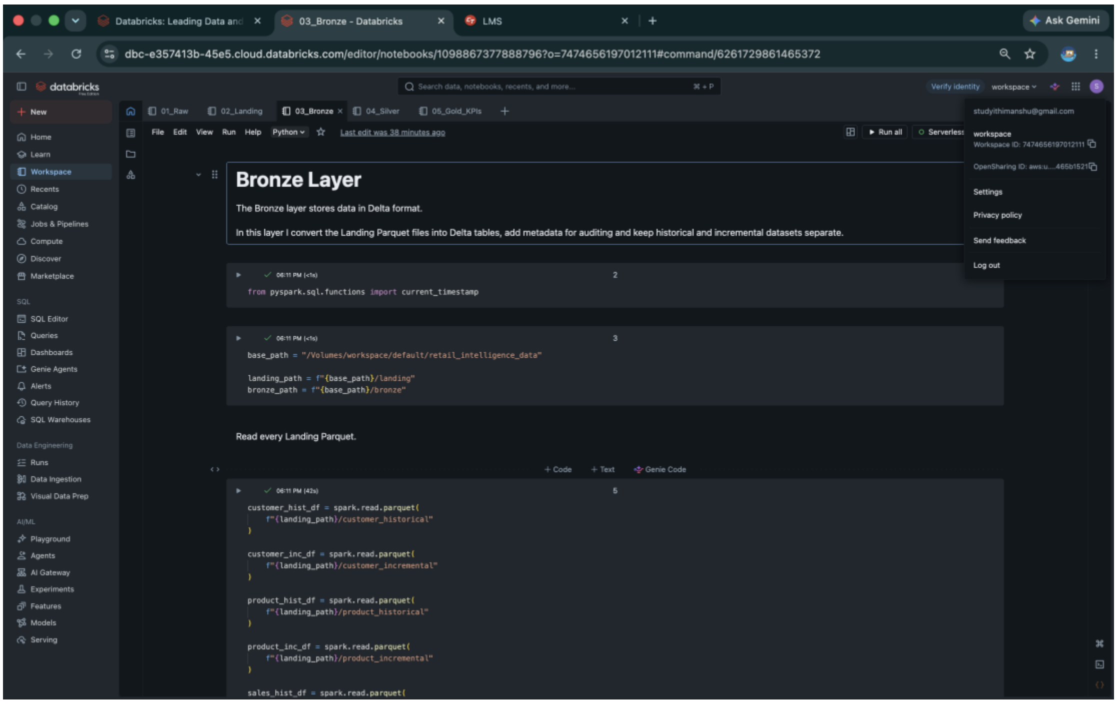
Validation checks are also performed to ensure that the number of records matches the audit files before moving further.



Bronze Layer

The Bronze layer converts Parquet files into Delta format.

Metadata such as ingestion timestamp is added while keeping historical and incremental datasets separate.

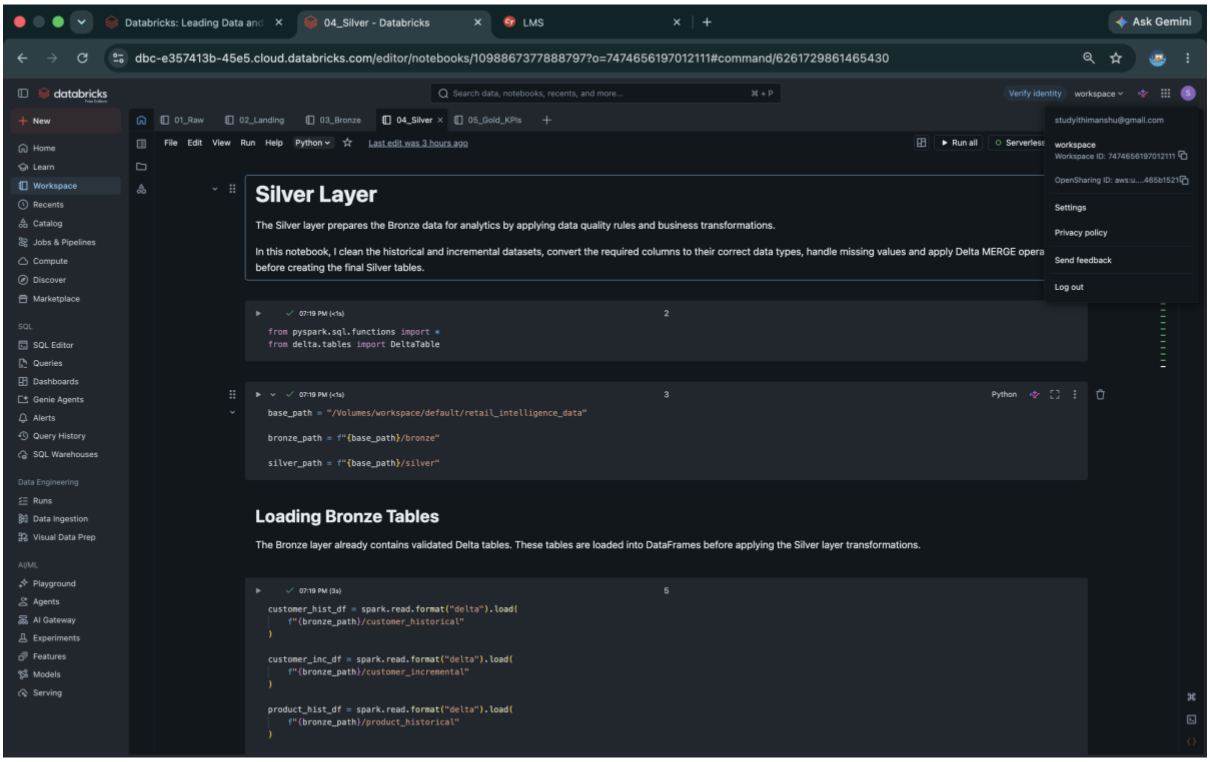


Silver Layer

The Silver layer performs all business transformations.

Activities performed:

- Handling missing values
- Standardising data
- Data type conversions
- Duplicate removal
- Business rule implementation
- Incremental customer processing using SCD Type 2



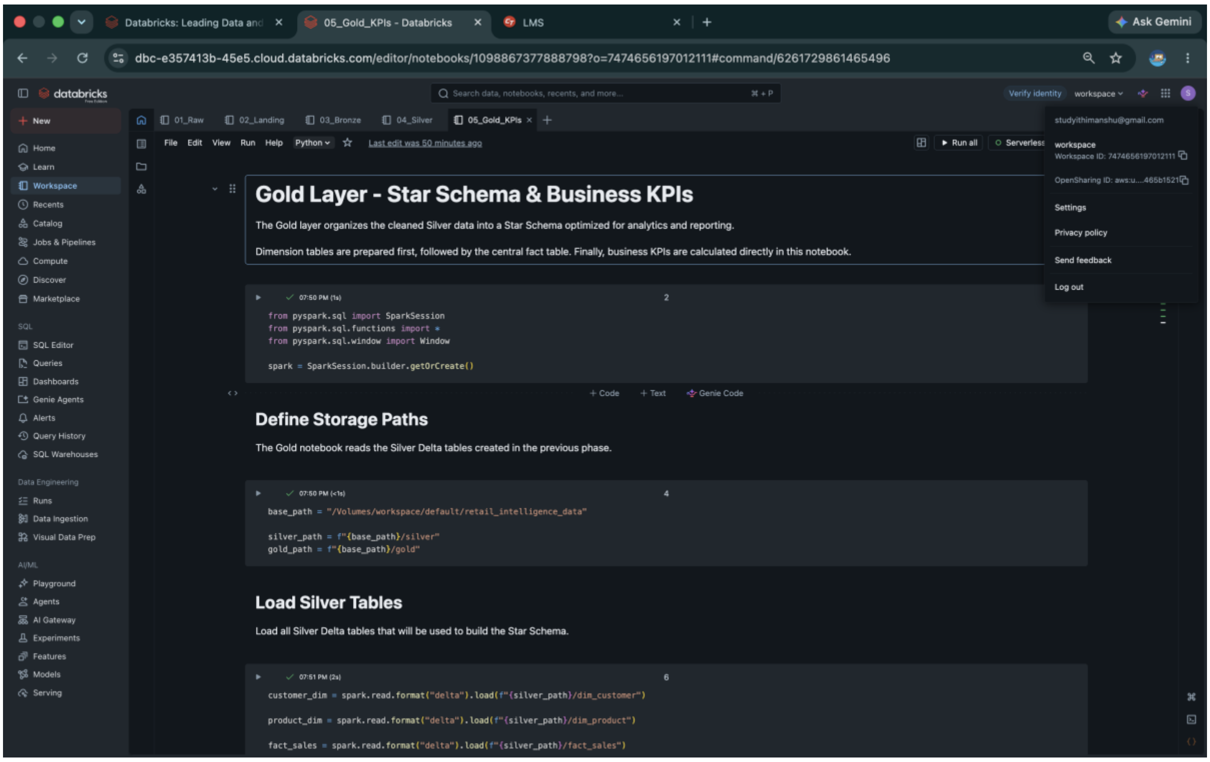
Gold Layer

The Gold layer creates analytical tables using the Star Schema model.

Tables created:

- Customer Dimension
- Product Dimension

- Date Dimension
- Promotion Dimension
- Sales Fact Table



Star Schema and Gold Layer

The Gold layer contains business-ready tables that are directly used for reporting and analytics.

The project creates four dimension tables and one fact table following the Star Schema design.

Dimension Tables

Customer Dimension

Stores customer-related information.

Product Dimension

Stores product details.

Date Dimension

Stores date attributes used for time-based analysis.

Promotion Dimension

Stores promotional campaign details.

Fact Table

Stores every sales transaction and references all dimension tables.

07:56 PM (3s)

```python
print("Customer Dimension :", customer_dim.count())
print("Product Dimension  :", product_dim.count())
print("Date Dimension     :", dim_date.count())
print("Promotion Dimension:", dim_promotion.count())
print("Fact Table         :", fact_sales_gold.count())
```

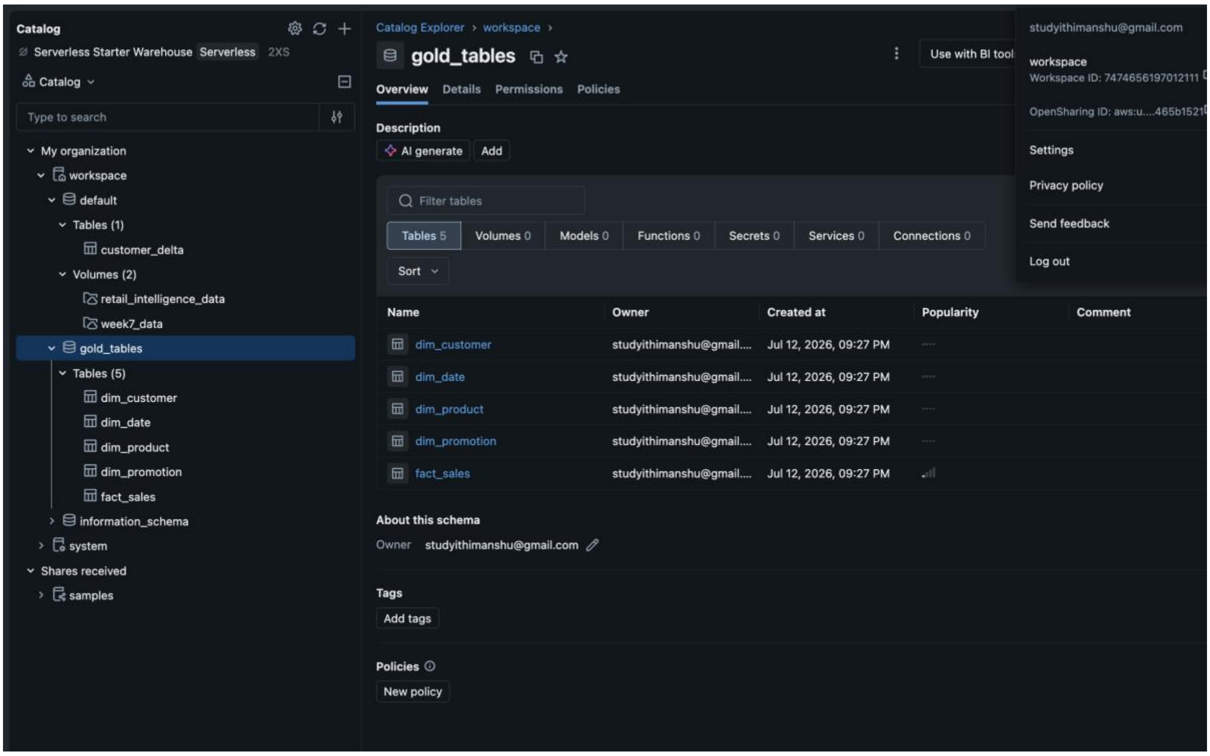
See performance (5)

```
Customer Dimension : 1050
Product Dimension  : 1041
Date Dimension     : 996
Promotion Dimension: 1570
Fact Table         : 2000
```

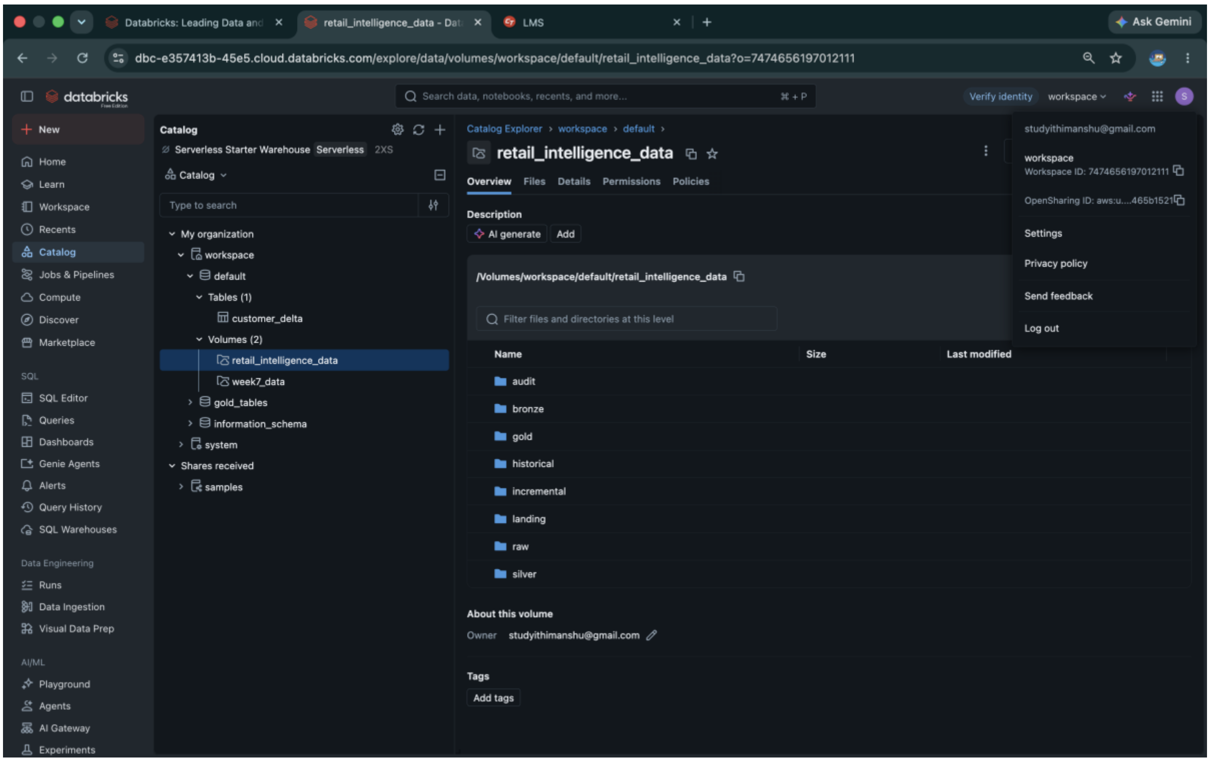
Features Implemented

- Incremental Data Processing
- SCD Type 2
- Delta Lake
- Star Schema
- Batch ETL Pipeline
- Business KPI Generation

Gold Tables



Catalog Registered Tables



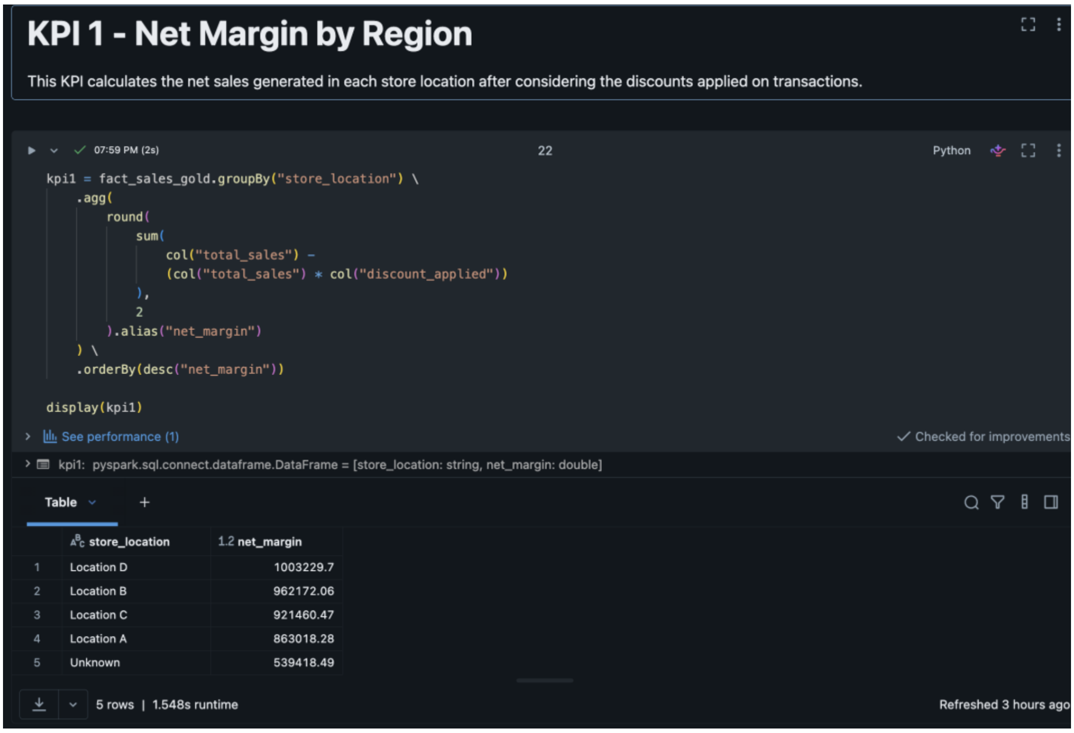
Business KPI Analysis

The Gold layer was used to generate business insights.

KPI 1

Net Margin by Region

This KPI calculates the net sales generated by each store location after considering the discounts applied on transactions. It helps identify the regions contributing the highest revenue.



KPI 1 - Net Margin by Region

This KPI calculates the net sales generated in each store location after considering the discounts applied on transactions.

```python
kpi1 = fact_sales_gold.groupBy("store_location") \
    .agg(
        round(
            sum(
                col("total_sales") -
                (col("total_sales") * col("discount_applied"))
            ),
            2
        ).alias("net_margin")
    ) \
    .orderBy(desc("net_margin"))

display(kpi1)
```

kpi1: pyspark.sql.connect.dataframe.DataFrame = [store_location: string, net_margin: double]

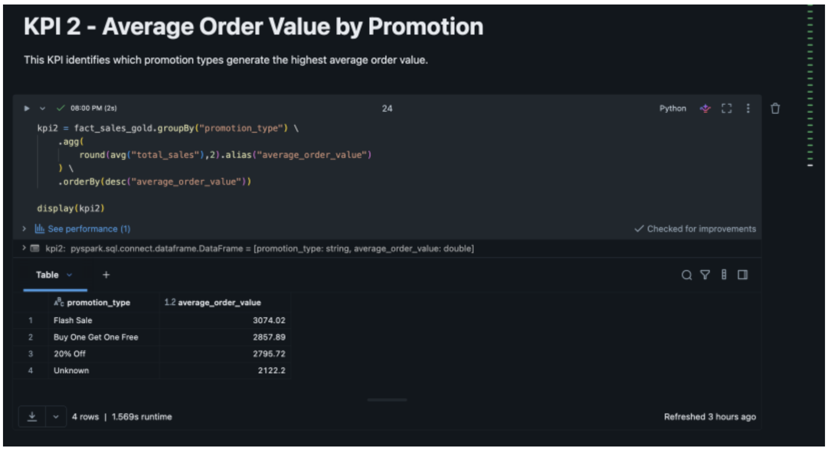
	store_location	net_margin
1	Location D	1003229.7
2	Location B	962172.06
3	Location C	921460.47
4	Location A	863018.28
5	Unknown	539418.49

5 rows | 1.548s runtime

KPI 2

Average Order Value by Promotion

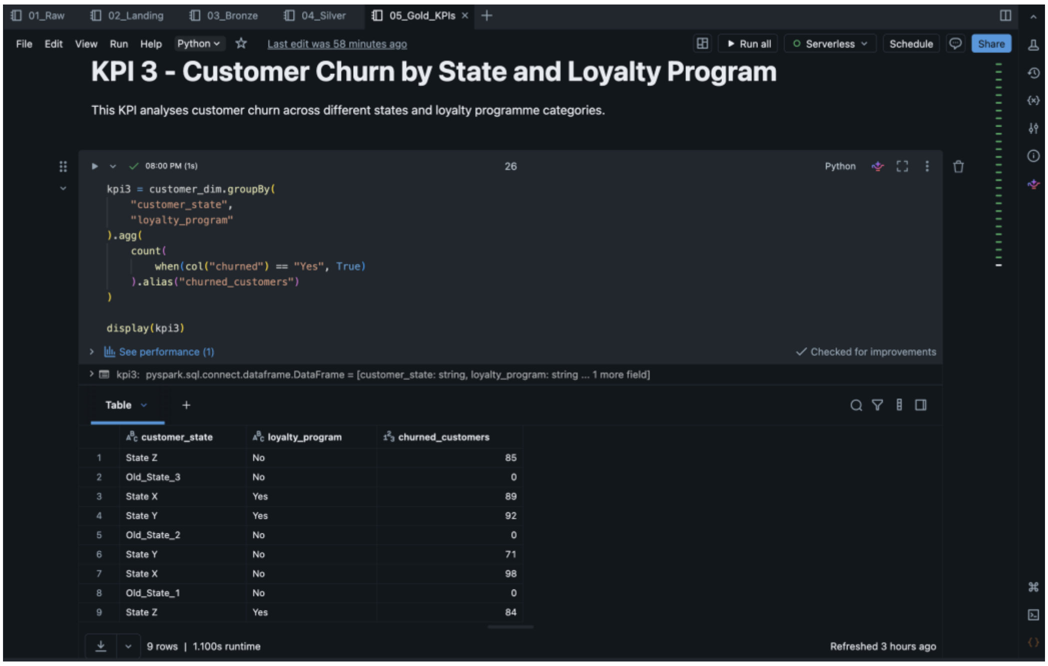
This KPI measures the average order value for different promotion types. It helps evaluate which promotional campaigns are more effective in generating higher sales.



KPI 3

Customer Churn by State and Loyalty Program

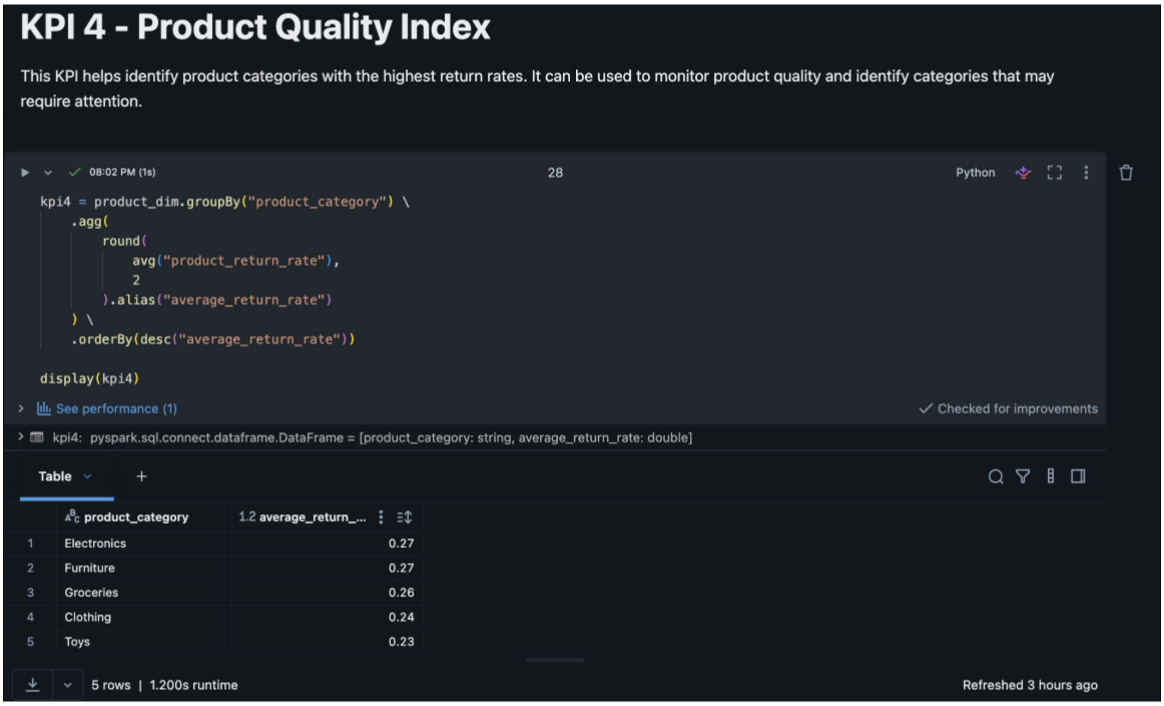
This KPI analyses customer churn across different states and loyalty program categories. It helps understand customer retention patterns and identify areas where churn is higher.



KPI 4

Product Quality Index

This KPI calculates the average product return rate for each product category. It helps identify product categories that may require quality improvements or further investigation.



KPI 4 - Product Quality Index

This KPI helps identify product categories with the highest return rates. It can be used to monitor product quality and identify categories that may require attention.

```python
kpi4 = product_dim.groupBy("product_category") \
    .agg(
        round(
            avg("product_return_rate"),
            2
        ).alias("average_return_rate")
    ) \
    .orderBy(desc("average_return_rate"))

display(kpi4)
```

kpi4: pyspark.sql.connect.dataframe.DataFrame = [product_category: string, average_return_rate: double]

	product_category	average_return_...
1	Electronics	0.27
2	Furniture	0.27
3	Groceries	0.26
4	Clothing	0.24
5	Toys	0.23

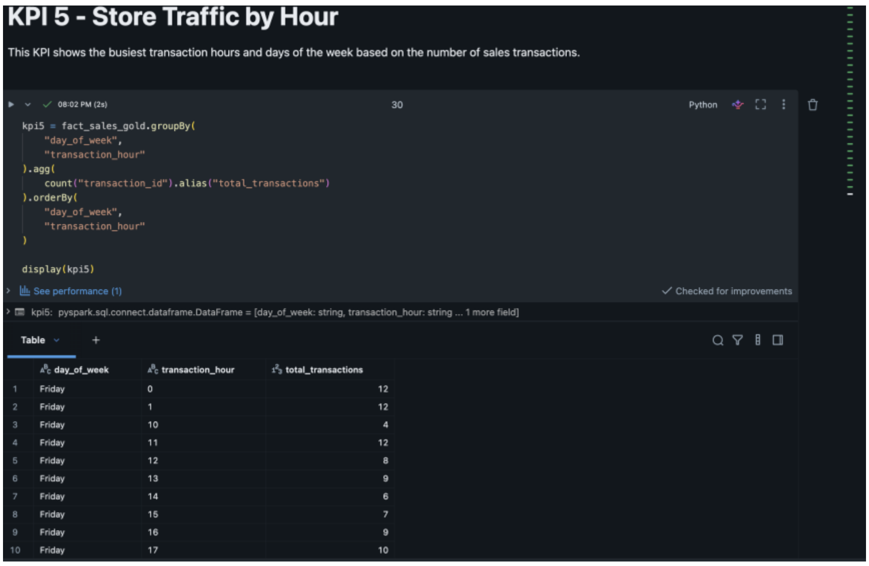
5 rows | 1.200s runtime

Refreshed 3 hours ago

KPI 5

Store Traffic by Hour

This KPI shows the number of transactions across different hours of the day and days of the week. It helps identify peak business hours and supports better staffing and operational planning.



KPI 5 - Store Traffic by Hour

This KPI shows the busiest transaction hours and days of the week based on the number of sales transactions.

```python
kpi5 = fact_sales_gold.groupBy(
    "day_of_week",
    "transaction_hour"
).agg(
    count("transaction_id").alias("total_transactions")
).orderBy(
    "day_of_week",
    "transaction_hour"
)

display(kpi5)
```

kpi5: pyspark.sql.connect.dataframe.DataFrame = [day_of_week: string, transaction_hour: string ... 1 more field]

	day_of_week	transaction_hour	total_transactions
1	Friday	0	12
2	Friday	1	12
3	Friday	10	4
4	Friday	11	12
5	Friday	12	8
6	Friday	13	9
7	Friday	14	6
8	Friday	15	7
9	Friday	16	9
10	Friday	17	10

Conclusion

This internship helped me understand the complete workflow involved in building a modern data engineering pipeline.

Starting from Python programming and SQL, I gradually learned data engineering concepts, Apache Spark, PySpark, Databricks, Delta Lake, ETL pipelines, and Star Schema modelling. The final Retail Intelligence project allowed me to apply these concepts in a practical scenario by processing historical and incremental retail datasets through the Medallion Architecture.

Working on this project also improved my understanding of data cleaning, batch processing, business transformations, and KPI generation. Overall, the internship strengthened both my technical knowledge and my confidence in building data engineering solutions.

Acknowledgement

I would like to sincerely thank **Celebal Technologies** for providing me with the opportunity to participate in this internship and gain practical exposure to modern data engineering tools and technologies.

I am especially grateful to **Yashasvi Ma'am** and **Raj Sir** for their continuous guidance, encouragement, and support throughout the internship. Their clear explanations, practical teaching approach, and constant motivation helped me understand complex concepts step by step and successfully complete the weekly assignments as well as the final Retail Intelligence project.

This internship has been an important milestone in my learning journey, and I am thankful for the knowledge, experience, and confidence I gained during these eight weeks.